

Design has long been about products and brand. In today's competitive marketplace, however, that perspective is too narrow. Design, now, must illuminate a broader and more strategic range of organizational activities.



Rachel Cooper, Professor,
Lancaster University



Sabine Junginger,
Lecturer, Lancaster
University



Thomas Lockwood,
President, DMI

Design Thinking and Design Management: A Research and Practice Perspective

by Rachel Cooper, Sabine Junginger, and Thomas Lockwood

As design professionals and design researchers, we feel the urge to reflect on what the recent focus on design thinking contributes to our understanding of design—and how it may inform design management. Design thinking is, of course, not a new concept or practice. It has been with and around us ever since there was design, conscious or unconscious. But some current interpretations offer new nuances, and that does have an impact on how we practice and theorize about design.

Our dictionaries distinguish between thinking of, thinking about, and thinking through. They equate the

activity of thinking of with imagining, visualizing, dreaming up. In contrast, to think about something is described as an activity during which one considers, reflects, and deliberates. Finally, to think through something is to understand, to grasp, to figure it out. Design thinking seems to encompass all three of these qualities.

Most of us would describe design thinking as a key activity. Yet, when we take a closer look, it is not always clear whether we are thinking about design, thinking of design, or thinking through design. It appears that each of these is a different kind of activity, with direct consequences for

the practice of designing—and the design of business itself. If we take these common definitions as a starting point, we might say that design thinking as the activity of thinking about design allows us to reflect on questions of who can design, and what can be designed. In this reflective mode, design thinking offers us ways to study the perceptions, expectations, and capabilities assigned to and associated with the theories and practices of designing. As such, design thinking offers opportunities to distinguish among particular design methods and design principles as they involve different foci (for

example, human-centered or technology-driven, product-centered or system-centered). These include, for example, participatory and collaborative design methods and visualization techniques for ideation and sense-making in two and three dimensions (that is, mapping and rapid prototyping). Thinking about design shifts the perspective from an isolated product to a more system-wide perspective. It is no coincidence that service design is becoming more of a focus here.

Building on the common definitions, to think of design then means to imagine, visualize, and dream up new understandings, new roles for

the design discourse. Most businesses are familiar with the first two kinds of design thinking. They think of design when they design specific products and services; they think about design as, perhaps, a tool for marketing. But now a new way to employ design thinking—thinking through design—is emerging, and it promises to have a much more profound impact on the way business itself is being conducted.

A wider focus

Design thinking applied to business strategy and business transformation involves the visualization of concepts

think like a designer? While many practicing designers still find themselves hard pressed to give an articulate answer, scholars in business and management have made some astute observations.

Roger Martin¹ has pointed to design's characteristics of integrative thinking and, in this sense, followed an earlier exploration by Peter Senge, whose Fifth Discipline emphasized the importance of system thinking. Senge's latest work, *The Necessary Revolution*, links even more with design thinking and design methods. Among design theorists, Richard Buchanan² brought the "wicked," or

Design thinking applied to business strategy and business transformation involves the visualization of concepts and the actual delivery of new products and services.

design, new practices, and new applications. What if an organization were a product? What will the profile of a designer look like in the future? How will this affect the profession and practice of design management?

While thinking about and thinking of design are core elements of design thinking that offer reflections on the past and, in some ways, the future of design, it is the idea of thinking through design that currently shapes

and the actual delivery of new products and services. The role of design is thus broader and more comprehensive than the role it is assigned in traditional product development. Increasingly synonymous with thinking like a designer, thinking through design has the greatest potential to establish the activities involved in designing as a core capability, one that goes beyond its traditional boundaries. But what exactly does it mean to

complex, nature of some design problems (that is, that they are difficult to solve because of incomplete, contradictory, or changing requirements) to our attention and highlighted the

1. Cf. Martin's *The Opposable Mind* (Boston: Harvard Business School Press, 2007).

2. See R. Buchanan, "Wicked Problems in Design Thinking," *Design Issues*, vol. 8, no. 2 (Spring 1992), and "Design as Inquiry: The Common, Future, and Current Ground of Design," paper presented at the DRS Futureground International Conference, Melbourne, Australia, 2004.

values of design inquiries and systems thinking. The interesting aspect here is that we get a pointer on how designing is actually done. We could elaborate on these aspects of design thinking for a full paper, but we feel a more urgent need to reflect on what the recent focus on design thinking has already achieved and contributed to the practice and theory of design management.

We should, however, in this context not forget the work and the people who laid the early foundations of what is now the area of design management. This list includes Bruce Archer, Michael Farr, James Pilditch,

Journal and the *Design Management Review*, a total of 850 articles dedicated to the problems and practices of design management. Along with 32 DMI case studies (now distributed by Harvard University), these works document our changing understanding of design thinking. In particular, we can detect a shift in our notion of the product and a shift in the tasks and challenges managers face.

The concept of design thinking is now taking hold in management and is paving the way for design to address new problems in the organization. With that, design thinking frees design activities from the product

current focus of design thinking, particularly as applied in the US, centers on innovations and business transformations that begin with people; it involves the discovery of unmet needs and opportunities, as well as the creation of new visions and alternative scenarios, that can reorient an organization around the people it serves. It is part of the fuzzy front end of idea generation and is also being adopted to solve “wicked” problems and help reinvent business.

On the other hand, design thinking has challenged us to inquire into the design process itself. At first glance, it appears that the current

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Mark Oakley, Peter Gorb, and Angela Dumas, among many others. Re-visiting some of these early works, one can often find a quest for design thinking and its role in the organization as the motivation for these authors. The DMI, as a non-profit organization, has played a lead role in advancing the ideas around design management. Founded by Bill Hannon in 1975, the DMI has since published 74 issues of the *Design Management*

and allows design thinking to be applied outside the traditional realms of design and to different kinds of problems (those of organization, strategy, mission, and so on). Indeed, the phrase *design thinking* has created excitement among many people previously untouched by design, and this has generated new opportunities for designers to engage with business, management, and other functions and levels within the organization. The

design and management discourse on design thinking contributes to our understanding of the design process. But this is not always the case. In fact, it is here that the potential problems and challenges for design thinking lie. Since design thinking and design methods always go hand in hand—that is, in design the thinking is informed by the doing, and vice versa—there is concern among some designers that the emphasis

on thinking might overshadow the importance of making. Some worry about splitting the design process into two separate activities, when design research is about the active engagement between the two. This concern might be unwarranted, as most current approaches to design thinking do involve design methods.

A more valid concern is the term's currency as a buzzword. While the global popularity of design as a business tool is undeniable, design thinking could well end up being the business-process flavor of the month, if design research fails to establish itself on its own terms. Design think-

Implications of design thinking for design management

Design management is the ongoing management—and leadership—of design organizations, design processes, and designed outcomes (which include products, services, communications, environments, and interactions). Generally, design management has been more concerned with individual design projects and with incremental developments and improvements, whereas design thinking represents a more radical shift in an organization's overall way of doing business. Design thinking addresses the fundamental assumptions,

and customer touch-points on an organizational level. Not surprisingly, this shift is accompanied by some anxiety about what design management is and is not, what it should or should not be. It is clear, however, that design thinking in all its guises has been a contributing factor in the activity of managing design and the delivery of new processes, products, and services.

One might view the rise of design thinking as a way forward for design in the organization, one that can only benefit the practice and research of design management. There will be a need for design management

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ing clearly coincides with the popularity of design at large, but there is more. One could point to influential management books, such as Malcolm Gladwell's *The Tipping Point*, Tom Kelley's *The Art of Innovation*, Richard Florida's *The Rise of the Creative Class*, and Daniel Pink's *A Whole New Mind*. All of these are relevant to the development of design thinking, or thinking like a designer, as a valid business tool.

values, norms, and beliefs that make an organization what it is. Design management, like most areas of design practice, is experiencing the shift from managing the design of tangible products to managing the design of innovation and services. This poses new problems for managing and for designing. There are some who see new opportunities as design moves from lower-level product-centered design strategies to the complexities involved in designing business processes

to serve different managerial and organizational purposes. We would like to argue that the rise in design thinking has helped to raise awareness of design management at different levels in the organization and, with that, has contributed to a clearer picture of design management. Indeed, the impact on how we perceive design management is enormous, if we consider, for example, recent writings on "managing as designing," which offer some of the most useful illustrations

of design thinking for organizations.³ Managing as designing can be viewed in some ways as a direct response to design management, which for some seems to have focused too much on “designing as managing.” This is, of course, an oversimplification, and much of the research conducted as part of design management will continue to inform these new practices. What we are likely to see is the emergence of a clearer picture of what design management can be in an organization in any given context.

3 See, for example, R. Boland and F. Collopy, *Managing as Designing* (Stanford, CA: Stanford University Press, 2004).

What can we learn from the DMI Education Conference?

In many ways, the two DMI conferences in Paris last spring provide the first evidence that this clarification is already happening. At the international DMI Education Conference, we could see the full spectrum of design management as it directly relates to design thinking. The papers and case studies presented there point to an increasing consensus that the challenges for a design manager include the ability to 1) explain the values and roles of design for a specific organizational setting; 2) define what design means for this organization; 3) com-

municate these roles internally and externally; and 4) create the space for design in this organization to assume and fulfill this role.

At the subsequent DMI professional conference, *Design as the Linking Force*, practicing design managers shared their views on design thinking as a means to strengthen the role of design in the organization. Both practitioners and researchers emphasized the ability to visualize concepts and ideas early on and to guide an emerging rather than deterministic inquiry as one of the key characteristics of design thinking.

Figure 1 reflects the different



Figure 1. Exploring design thinking in design management: Participants of the 2009 International DMI Education Conference Roundtable Session mapped their ideas, concepts, practices, and theories as part of a discussion on design thinking. DMI President Tom Lockwood moderated the session.

interpretations of design thinking evident in the presentations given at the DMI Education Conference. This mass brainstorm of the entire audience of delegates illustrates how broad is the notion of design thinking, and as such reflects the need to further define that notion, or indeed categorize it as we have done above.

It also clearly illustrates a progression or shift in terms of our understanding and implementation of design thinking. For instance (see Figure 2), in regions where design management is just emerging, researchers tend to follow classic design management approaches based on products, technology, and brand. In regions where

design management has a tradition, academic research begins to inform these practices, complemented by professional reflections. However, the design thinking underlying these design management practices and research tends to remain product-centric. In turn, this begins to change in regions where design has been es-

MAKING SENSE OF CURRENT DESIGN MANAGEMENT RESEARCH & PRACTICE

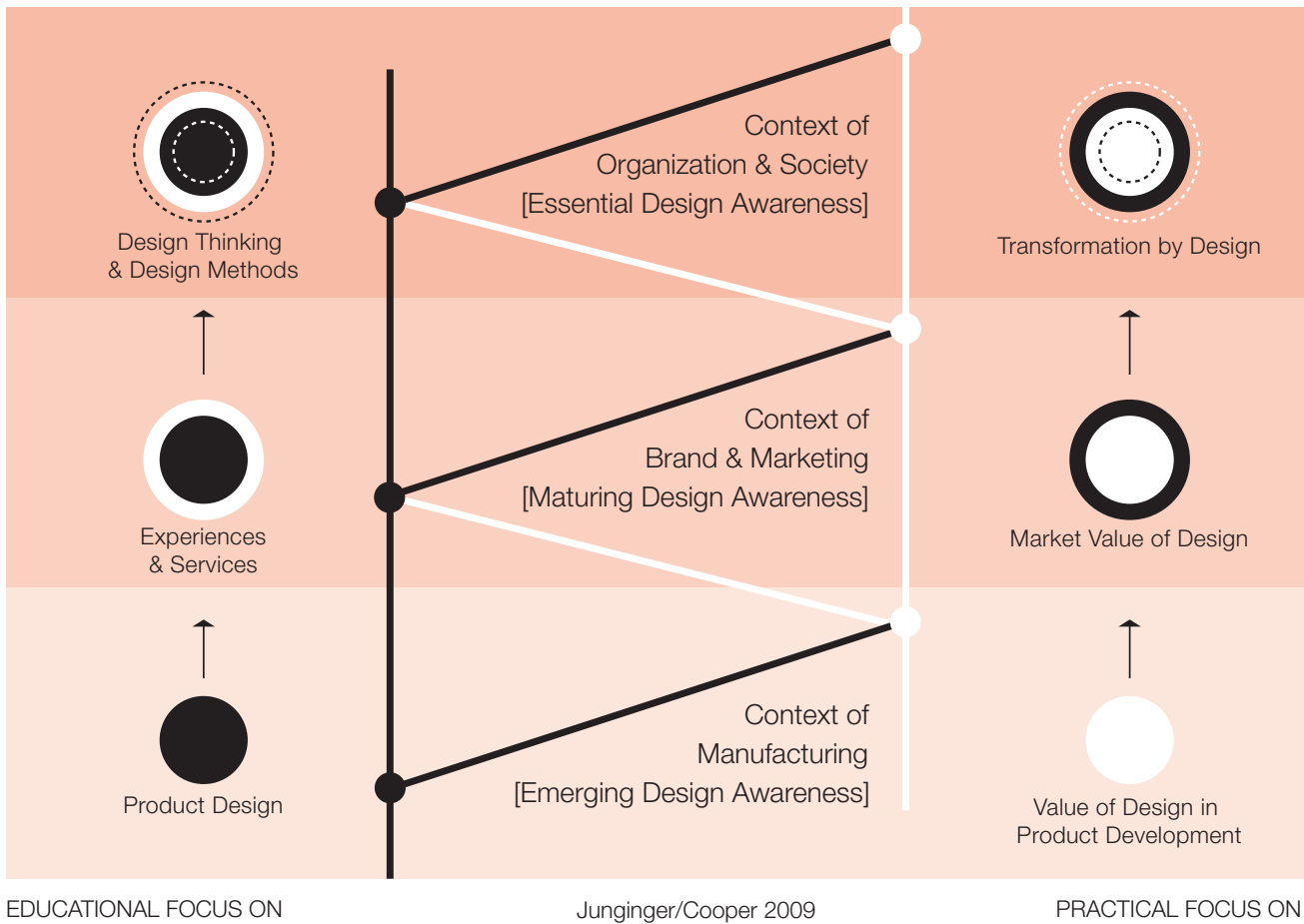


Figure 2. Contributions to the 2008 DMI Education Conference in Paris reflected the two research areas of design management by bringing together both academic and professional insights. At the lower rungs of the ladder, research tends to center on traditional design practices and the study of products and brand. Further up the ladder, system theories, non-traditional products, and thus, thinking through design are being explored.

established as part of an organization. Here we can see how design thinking is freeing itself from these previous traditions and emerging as a practice independent of the product. Instead, design thinking now focuses on the characteristics of a problem that needs solving.

Conclusion/Summary

If we consider design thinking as an activity that is key to developing products, then design thinking has been around ever since design has been around. What is different today

is that it has been discovered by more people as a valuable tool with which to address problems and issues that do not necessarily involve a product to manufacture for sale. There are multiple ways in which design practitioners and design researchers have made use of the term. It appears that design thinking provides managers with a new way to think of new products and services and about ways of designing them. The imperative now is for business schools and design schools alike to open new paths for students to acquire design skills that

allow them to think through design. This involves the ability to quickly visualize problems and concepts, the development of people-based scenarios, and the design of business strategies based on design research methods. In this new way, managers are able to see how design methods and design principles can help them navigate the uncertainties and complexities they now face. Moving beyond its traditional role, design is establishing itself as a tool with which to address a wide range of issues, from strategy to social change.

The Progression of Design Management

Stage 1. Design management in the context of manufacturing

The first and most obvious context for the practice and research of design management is that of the manufacturing organization. In fact, it is here that design management originated in the 1960s. The main purpose of a product here is to succeed in a competitive market. In this context, design management concerns itself with management issues that directly relate to the product development process. Areas of research, therefore, include the search for best practices in product development processes; relationships among product designers, engineers, and marketing experts (that is, integrated product development), and also the role of the designer in a product development team. Design management researchers who study these areas tend to focus on the manufacturing process. Design in this context is often understood as adding value, which poses questions such as: What value does design add to a manufactured good? When does design add value in the development process? How can this value be measured?

Only a few books and articles on design management have been (officially) translated into languages of countries where design management is now emerging. This makes it difficult for design researchers (and practitioners) in countries where design management is a new field to access existing and current literature and case studies. Despite a rich history of national and cultural arts, some countries have just recently begun to invest in systematic design education both in theory and practice. Their graduates are geared toward the needs of the business landscape surrounding them, where design is mainly understood as product design. While one might think of Asia or Russia as examples here, product design continues to be the justification for design management for many SMEs in Western European countries, even though design practice and education have a longer history there. This shows that research into design management in the manufacturing context remains relevant. Yet, as the role of manufacturing shrinks, we can also see why new avenues are being explored in design management practice and research.

Stage 2. Design management in the context of marketing and branding

The second major context for the practice and research of design management builds on the role of the product in the manufacturing context. Thus, the product continues to be understood as mostly a good for sale, and its purpose remains that of succeeding in a competitive market. However, the problems of marketing and branding broaden the scope and add new questions that relate to the perception and experience of a product by the people defined as the target group. It is no coincidence that the role of design management in branding and marketing evolved earlier in countries in which manufacturing now plays a significantly lower role in the national economy than it did even 10 years ago. This holds true for many Western European countries and North America.

It is tempting to consider only countries that have established traditions in design theories and practices. Many of these countries are indeed home to internationally renowned and admired brands (for example, BMW, Audi, Apple, Starbucks). However, some countries, especially in Asia, are investing heavily in the development of an overall design research infrastructure around branding and marketing. There is unprecedented support and synergy on a national level in the form of design policies and design education programs, both of which are informed by continuous international exchanges and engagements.

Design management research in the context of marketing and branding is predominantly centered on a tangible product and/or its accompanying services. It is in this area of practice and research that the concept of service design took shape initially (although each country interprets service in its own terms). This product-centeredness contributes to a persistent perception that design itself is merely offering a service to the process of manufacturing goods for sale. Indicative of the situation is the fact that in many of these countries, design research itself is not yet established as a recognized field. We might think of these as countries in which design continues to be taught in “applied universities” (product development being considered a practical task), which by local law are not allowed to supervise doctoral research or to grant doctoral degrees.

Stage 3. Design management in the context of the organization and society

The third, and fairly new, context for the practice and research of design management is that of the organization and of society itself. Where design thinking and design activities are now established parts of an organization, either in manufacturing or in marketing and branding, design management is changing its course from one of designing as managing to one of managing as designing. Here we can see how design thinking is freeing itself from these previous traditions and emerging as a practice independent of the product. In this context, design thinking focuses on the characteristics of a problem. However, because of the lack of infrastructure for design research and the difficulty with which traditional design programs might move beyond the product, we see a lot of research happening in management and business schools. Witness, for example, the rapid developments in North American business schools, which are looking for alternative, economically sustainable business models and business practices. However, instances of thinking through design are being explored in other places, too—most notably The Netherlands and the UK, whose Design Council has been very active. Design research, in general, is more advanced in these countries, with established doctoral programs and an increasingly accessible body of research funding design education. Design research has also been accepted as a valuable contributor to research across scientific domains; many design scholars are working with scholars in science, social science, and humanities to address social, environmental, political, and economic questions. There are currently efforts on an EU level to create a stronger research base for design and the role of design in innovation that goes beyond the product to cover the entire system is being explored.

Even given those basic facts, we can find representatives of each stage of design thinking in pretty much every country. The difference from country to country may well lie in the prevalence of design researchers pursuing these levels individually, as opposed to a more systematic framework in which design thinking can be explored and discussed.

Suggested Readings

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by Will Ayres & Scott Lerman

MAYBE IF I JUST LOSE THE TIE?